

Name: _____

ID: _____

Section: _____

Q1: Given data (x,y): (1,2),(2,3),(3,5)

a) Compute the least squares line $y = \beta_0 + \beta_1 x$ by normal equations.

Given data

$$(x, y) : (1, 2), (2, 3), (3, 5)$$

(a) Least Squares Line using Normal Equations

Step 1: Compute means

$$\bar{x} = \frac{1+2+3}{3} = 2, \quad \bar{y} = \frac{2+3+5}{3} = \frac{10}{3} = 3.33$$

Step 2: Compute slope β_1

$$\beta_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

x	y	$x - \bar{x}$	$y - \bar{y}$	Product
1	2	-1	-1.33	1.33
2	3	0	-0.33	0
3	5	1	1.67	1.67

$$\sum (x - \bar{x})(y - \bar{y}) = 3, \quad \sum (x - \bar{x})^2 = 2$$

$$\beta_1 = \frac{3}{2} = 1.5$$

Step 3: Compute intercept β_0

$$\beta_0 = \bar{y} - \beta_1 \bar{x} = 3.33 - (1.5)(2) = 0.33$$

Least Squares Line

$$\hat{y} = 0.33 + 1.5x$$

b) Compute Total Sum of Squares (SST), SSE — Error / Residual Variation, Residual Sum of Squares (SSR) Explained by the Model and then R^2 .

(b) SST, SSE, SSR and R^2

Step 1: Predicted values

x	y	$\hat{y}$
1	2	1.83
2	3	3.33
3	5	4.83

Total Sum of Squares (SST)

$$SST = \sum (y - \bar{y})^2$$

$$= (-1.33)^2 + (-0.33)^2 + (1.67)^2 = 4.67$$

Residual Sum of Squares (SSE) (Error Variation)

$$SSE = \sum (y - \hat{y})^2$$

$$= (0.17)^2 + (-0.33)^2 + (0.17)^2 \approx 0.17$$

Regression Sum of Squares (SSR) (Explained by Model)

$$SSR = SST - SSE = 4.67 - 0.17 = 4.50$$

Coefficient of Determination

$$R^2 = \frac{SSR}{SST} = \frac{4.50}{4.67} \approx \boxed{0.964}$$

About 96.4% of the variation in y is explained by the model.

c) If one more observation (10,20) is appended, re-compute R^2 . Explain why R^2 increased or decreased even if the fitted slope barely changed.

(c) Add observation (10, 20) and re-compute R^2

New means

$$\bar{x} = \frac{1+2+3+10}{4} = 4, \quad \bar{y} = \frac{2+3+5+20}{4} = 7.5$$

Re-computing regression gives approximately:

$$\hat{y} \approx -0.74 + 2.06x$$

The new $R^2 \approx 0.99$ (very high).

R^2 can increase simply by adding a well-aligned extreme point, even if the regression line changes very little.

Q2: You fit linear regression and obtain residuals e_i . For a 3-point dataset residuals are $e=[1,-2,1]$ and $h_i=[0.2,0.6,0.2]$ (leverage values).

a) Compute the standardized residuals.

Given:

- Residuals

$$e = [1, -2, 1]$$

- Leverage values

$$h = [0.2, 0.6, 0.2]$$

- Number of observations $n = 3$
- Simple linear regression $\Rightarrow$ number of parameters $p = 2$

(a) Standardized Residuals

Formula

$$r_i = \frac{e_i}{\sqrt{MSE(1 - h_i)}}$$

Step 1: Compute Mean Squared Error (MSE)

$$SSE = \sum e_i^2 = 1^2 + (-2)^2 + 1^2 = 6$$

$$MSE = \frac{SSE}{n - p} = \frac{6}{3 - 2} = 6$$

Step 2: Compute standardized residuals

Point 1

$$r_1 = \frac{1}{\sqrt{6(1 - 0.2)}} = \frac{1}{\sqrt{4.8}} \approx 0.46$$

Point 2

$$r_2 = \frac{-2}{\sqrt{6(1 - 0.6)}} = \frac{-2}{\sqrt{2.4}} \approx -1.29$$

Point 3

$$r_3 = \frac{1}{\sqrt{6(1 - 0.2)}} = \frac{1}{\sqrt{4.8}} \approx 0.46$$

Standardized Residuals

$$r \approx [0.46, -1.29, 0.46]$$

b) Identify which point is most influential using standardized residuals and explain how leverage changed interpretation.

Most Influential Point

Point 2 is the most influential

Raw residuals alone suggest point 2 is extreme $e^2 = -2$. **High leverage (0.6)** increases the denominator. This **reduces the standardized residual magnitude**. The large error is *partly expected* because the point lies far from the center of x . Standardized residuals adjust raw errors for leverage, preventing high-leverage points from being misclassified as outliers while still identifying them as influential.

Aspect	Without Leverage	With Leverage
Error size	Looks very large	Moderated
Influence	Overstated	Correctly adjusted
Conclusion	Outlier	High-leverage influential point

Q3: Given two models fit on same test set ($n=4$):

Model 1 residuals: [2, -2, 2, -2]

Model 2 residuals: [1, 1, -1, -1]

a) Compute SSE, MSE, RMSE for both models and rank which is better.

Given test set size

$$n = 4$$

Residuals:

- Model 1: [2, -2, 2, -2]
- Model 2: [1, 1, -1, -1]

(a) SSE, MSE, RMSE and Model Ranking

Formulas

$$SSE = \sum e_i^2, \quad MSE = \frac{SSE}{n}, \quad RMSE = \sqrt{MSE}$$

Model 1

$$SSE_1 = 2^2 + (-2)^2 + 2^2 + (-2)^2 = 16$$

$$MSE_1 = \frac{16}{4} = 4$$

$$RMSE_1 = \sqrt{4} = \boxed{2}$$

Model 2

$$SSE_2 = 1^2 + 1^2 + (-1)^2 + (-1)^2 = 4$$

$$MSE_2 = \frac{4}{4} = 1$$

$$RMSE_2 = \sqrt{1} = \boxed{1}$$

Model Ranking

Model 2 is better (lower SSE, MSE, and RMSE)

b) Same residuals, but target y values were recorded in thousands for Model 2 — explain the scaling effect on RMSE vs relative error metrics like R^2

New residuals:

$$[1000, 1000, -1000, -1000]$$

New RMSE

1. Square:

$$1000^2 = 1,000,000$$

2. Average:

$$MSE = \frac{4 \times 1,000,000}{4} = 1,000,000$$

3. Square root:

$$RMSE = \sqrt{1,000,000} = \boxed{1000}$$

Before

After (thousands)

RMSE = 1

RMSE = 1000

RMSE increased only because the unit changed, not because the model got worse.

Formula reminder

$$R^2 = 1 - \frac{SSE}{SST}$$

Now think simply:

- If you multiply all **y values** by 1000:
 - Errors become 1000× bigger
 - Variations also become 1000× bigger

So:

- SSE becomes 1000^2 times larger
- SST also becomes 1000^2 times larger

The ratio:

$$\frac{SSE}{SST}$$

Therefore **R^2 stays the same**

Suppose:

Model explains 90% of variation

If marks are:

out of 100 → still 90%

out of 100,000 → still 90%

Percentages don't care about units Absolute errors do

Metric	Depends on Units?	Why
RMSE	Yes	Measures error size
MSE	Yes	Squared error size
SSE	Yes	Total error
R^2	No	Ratio of explained variation

Q4: You have the dataset:

X	Y
1	2
2	5
3	10
4	17

a. Fit a linear regression: $Y = a + bX$.

(a) Linear Regression: $Y = a + bX$

Step 1: Compute means

$$\bar{X} = \frac{1+2+3+4}{4} = 2.5$$

$$\bar{Y} = \frac{2+5+10+17}{4} = 8.5$$

Step 2: Compute slope b

$$b = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sum(X - \bar{X})^2}$$

X	Y	X-2.5	Y-8.5	Product
1	2	-1.5	-6.5	9.75
2	5	-0.5	-3.5	1.75
3	10	0.5	1.5	0.75
4	17	1.5	8.5	12.75

$$\sum(X - \bar{X})(Y - \bar{Y}) = 25$$

$$\sum(X - \bar{X})^2 = 5$$

$$b = \frac{25}{5} = 5$$

Step 3: Compute intercept a

$$a = \bar{Y} - b\bar{X} = 8.5 - 5(2.5) = -4$$

✓ Linear Regression Model

$$Y = -4 + 5X$$

b. Fit a polynomial regression model $Y = b_0 + b_1X + b_2X^2$ and calculate the coefficients using the least squares method.

(b) Polynomial Regression: $Y = b_0 + b_1X + b_2X^2$

Observe the pattern in data:

X	Y	X^2
1	2	1
2	5	4
3	10	9
4	17	16

Notice:

$$Y = X^2 + 1$$

Let's verify quickly:

- $1^2 + 1 = 2$ ✓
- $2^2 + 1 = 5$ ✓
- $3^2 + 1 = 10$ ✓
- $4^2 + 1 = 17$ ✓

✓ Polynomial Regression Model

$$Y = 1 + 0X + 1X^2$$

So:

- $b_0 = 1$
- $b_1 = 0$
- $b_2 = 1$

This model fits **perfectly** using least squares.

c. Compute predictions for $X=3$ and compare errors. Which model fits better?

(c) Prediction at $X = 3$ and Error Comparison

True value:

$$Y = 10$$

Linear Model Prediction

$$\hat{Y} = -4 + 5(3) = 11$$

Error:

$$|10 - 11| = 1$$

Polynomial Model Prediction

$$\hat{Y} = 1 + 3^2 = 10$$

Error:

$$|10 - 10| = 0$$

✓ Which model fits better?

Model	Prediction	Error
Linear	11	1
Polynomial	10	0

Polynomial regression fits better

The polynomial regression model fits better because the relationship between X and Y is quadratic, resulting in zero prediction error compared to the linear model.

The data follows a quadratic pattern

Linear regression cannot capture this trend

Polynomial regression captures the true relationship exactly